SnipShot (Reel Studio): Flagship Architecture & Technical Blueprint

Document Version: 2.0.0
Status: Canonical System Architecture & Business Model Blueprint
Classification: Local-First Multimodal Pedagogical Video Studio (SGPC)
Target Platform: Linux / macOS / Windows (Tier 1–3 CPU & GPU Profiles)

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1. Product Thesis & The 'Local-First Studio' Business Model

A. The Core Economic & Architectural Paradox

Modern video editors, technical instructors, and coaching institutes (such as TechSummitTrainers) already own powerful, multi-core workstation laptops equipped with 14–18 CPU threads, 16GB+ RAM, and neural accelerators or GPUs. Yet, the current generation of automated clipping tools (`OpusClip`, `Vizard.ai`, `Descript`, `Klap.app`) forces them into an absurd operational model:

1. **Bandwidth Tax:** Uploading 90-minute 1080p/4K lecture recordings (3GB–8GB) over residential or campus internet to remote cloud servers.
2. **Recurring Per-Minute Cloud Tax:** Paying monthly subscriptions that ration video minutes with aggressive credit caps (e.g., $30–$80/month for only 150–300 minutes of exported video), where every timeline re-trim burns billable cloud credits.
3. **Privacy & Security Risk:** Pushing unreleased proprietary course curricula, student Zoom interactions, and exposed terminal credentials (`.env` files, API keys, database URLs) to third-party cloud multi-tenant storage.
4. **Pedagogical Failure:** Receiving generic face-centred crops that chop source code lines in half horizontally and output captions like `"use a fact"` or `"auto wired"`.

B. The Local-First Perpetual Paradigm Shift

Professional video production tools disrupted software by refusing to charge monthly per-minute cloud rents, instead delivering a professional, hardware-accelerated local engine with an unbeatable community tier and a one-time perpetual Studio license.

SnipShot applies this exact local-first paradigm to AI-powered educational short-form video creation:

- **Zero Cloud Cost:** 100% of the speech recognition (`faster-whisper`), screen OCR (`RapidOCR`), syntax parsing (`tree-sitter`), and video composition (`Pillow + FFmpeg`) runs locally on the user's host machine.
- **Instant Turnaround:** Re-trimming timeline boundaries or switching Brand Kits re-renders a 45-second 1080×1920 Reel in **6–12 seconds** via multi-threaded CPU pipes, without waiting in a remote cloud GPU queue.
- **Total Air-Gapped Privacy:** Ingested media, extracted student interactions, and codebases never leave `localhost`.

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|                                   BUSINESS MODEL COMPARISON
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|  Traditional Cloud Clipper SaaS (OpusClip / Vizard)       SnipShot (Local-First Studio
Engine)        |
|  ------------------------------------------------       ---------------------------------
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|  - Cloud Inference Cost: $0.12 - $0.25 / minute           - Cloud Inference Cost: ₹0.00
($0.00)       |
|  - Gross Margins: 35% - 55% (Server heavy)                - Gross Margins: 92% - 96%
(Software pure)  |
|  - Pricing: $30 - $120/mo recurring per editor            - Pricing: Hybrid Perpetual
Studio License  |
|  - Video Privacy: Cloud Upload Required                   - Video Privacy: 100% On-Device
Air-Gapped  |
|  - Re-render Time: 45s - 180s (Network + Queue)           - Re-render Time: 6s - 12s (Local
Pipe)     |
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C. Hybrid Perpetual Commercial Architecture

SnipShot operates on a three-tier commercial structure designed for institutes and solo creator-engineers:

1. **Community Free Tier (100% Local, Forever Free):**
   - Unlimited 1080×1920 exports with standard watermark.
   - Core `ScreenLex-ASR` vocabulary biasing and `SplitStage` Dual-ROI reframing.
   - Standard Brand Kit presets. Zero cloud dependencies.
2. **Studio Perpetual License (One-Time Payment — ₹14,999 / $199):**
   - Commercial license with watermark removal.
   - Full 10-Weapon Arsenal unlocked: `SafeCohort` secret redaction, custom institute Brand Kits, Outro QR code generation, and 4K/60fps rendering.
   - Lifetime local software access with 12 months of core algorithmic updates.
3. **Optional Annual Model & Template Pass (₹3,999/yr / $49/yr):**
   - Access to newly tuned Hinglish domain glossaries (AI/ML, DevOps, Rust, Web3).
   - Seasonal trend templates and viral hook archetype updates.
4. **Institute Enterprise Fleet Tier:**
   - Multi-seat volume licensing for coaching academies (TechSummitTrainers, CodeCampus) with centralized brand asset sync and on-premise OBS/Zoom batch folder watchers.

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2. Deep Competitor Technical Comparison Matrix

The table below contrasts **SnipShot (SGPC)** with leading commercial clippers and open-source baselines across 10 mission-critical technical dimensions for technical education:

Technical Dimension	OpusClip (`ClipAnything`)	Vizard.ai	Klap.app	Descript	OS Auto-Clipper (`MoneyPrinterTurbo`)	SnipShot (SGPC 2.0)
1. Hinglish & Code-Switch ASR	Weak (separates Hindi/English; mangles mixed romanized grammar)	Weak (Whisper base without domain prompt tuning)	Poor (drops mixed clauses)	Moderate (monolingual English focus)	Poor (raw Whisper baseline)	**Superior (ScreenLex-ASR):** Dynamically injects top 60 screen identifiers into `faster-whisper hotwords` + Romanized Hinglish prompt seed.
2. Technical Identifier Repair & Visual Proof	None (manual glossary only)	None (manual find & replace)	None	Partial (custom dictionary prompt)	None	**Autonomous (Double-Metaphone + Levenshtein):** Merges phonetic n-grams (`"use a fact"` → `useEffect`) with clickable IDE crop proof badges.

Technical Dimension	OpusClip (`ClipAnything`)	Vizard.ai	Klap.app	Descript	OS Auto-Clipper (`MoneyPrinterTurbo`)	SnipShot (SGPC 2.0)
3. 9:16 Code Legibility & Reframing	Naive face crop or letterbox (crops wide code lines)	Letterbox stack (shrinks font below 6pt)	Face crop (fails on screen shares)	Manual canvas resize / PiP	Fixed centre crop (`ih*9/16:ih`)	**Autonomous (SplitStage + Readable916):** Lifts wide lines $> 850\text{px}$ into 32px syntax-highlighted code cards; zero character truncation.
4. Left/Right IDE + Terminal Split	Crops out terminal or slides erratically	Requires manual dual-window cropping	Incompatible	Manual layer masking	Cuts terminal completely	**Automated (Dual-ROI SplitStage):** Detects bimodal spatial activity, transposing horizontal split into a Top-60% IDE / Bottom-40% Terminal vertical stack.
5. Boundary Completeness & Deixis	Viral hooks only (frequently starts on *"And this..."*)	Transcript hook keywords only	Hook heuristics	Semantic chaptering	Fixed duration cuts	**ClosedThought Triad Scorer:** Snaps strictly to grammatical sentence ends; penalizes dangling pronouns (*"isko"*, *"ye wala"*); enforces Bug2Fix closure.
6. Mode B Prompt-to-Code Explainer	Unavailable	Unavailable	Unavailable	Text-only script generation	Stock B-roll + TTS avatar	**VerifiedCodeReel:** Local `Qwen2.5-Coder` + `tree-sitter` syntax gate with audio-bookmark locked typing animations.
7. Dead-Air 2.5x Typing Hyperlapse	Cuts silence completely (destroys typing context)	Raw silence cut	Raw silence cut	Silence removal (audio cut only)	None	**Typing Hyperlapse:** Detects silent typing intervals $> 1.2\text{s}$ and speeds them up 2.5x during frame synthesis to eliminate dead air while preserving visual proof.
8. `.env` Secret & PII Auto-Redaction	None (cloud upload exposes all secrets)	None	None	None	None	**SafeCohort Shield:** Autonomous regex & Shannon

Technical Dimension	OpusClip (`ClipAnything`)	Vizard.ai	Klap.app	Descript	OS Auto-Clipper (`MoneyPrinterTurbo`)	SnipShot (SGPC 2.0)
						entropy scanner on screen OCR bounding boxes, applying localized Gaussian blur to secrets (`sk-`, API keys).
9. Screen-Grounded SEO & Outro QR Code	Generic LLM tags from audio transcript	Generic tags	Generic tags	None	Generic audio tags	**Growth Pack + Scan-to-Copy:** Generates tiered hashtags directly from AST identifiers and stamps a 'Scan-to-Copy-Code' QR code on the outro bumper.
10. Execution & Pricing Model	100% Cloud SaaS ($30–$80/mo recurring)	100% Cloud SaaS ($40–$90/mo recurring)	100% Cloud SaaS ($30/mo)	Cloud SaaS ($24–$40/mo)	Free (local script, but fails on code)	**100% Local-First Engine (₹0.00 cloud cost);** Perpetual license with zero marginal server expense.

3. The 10-Weapon Championship Arsenal

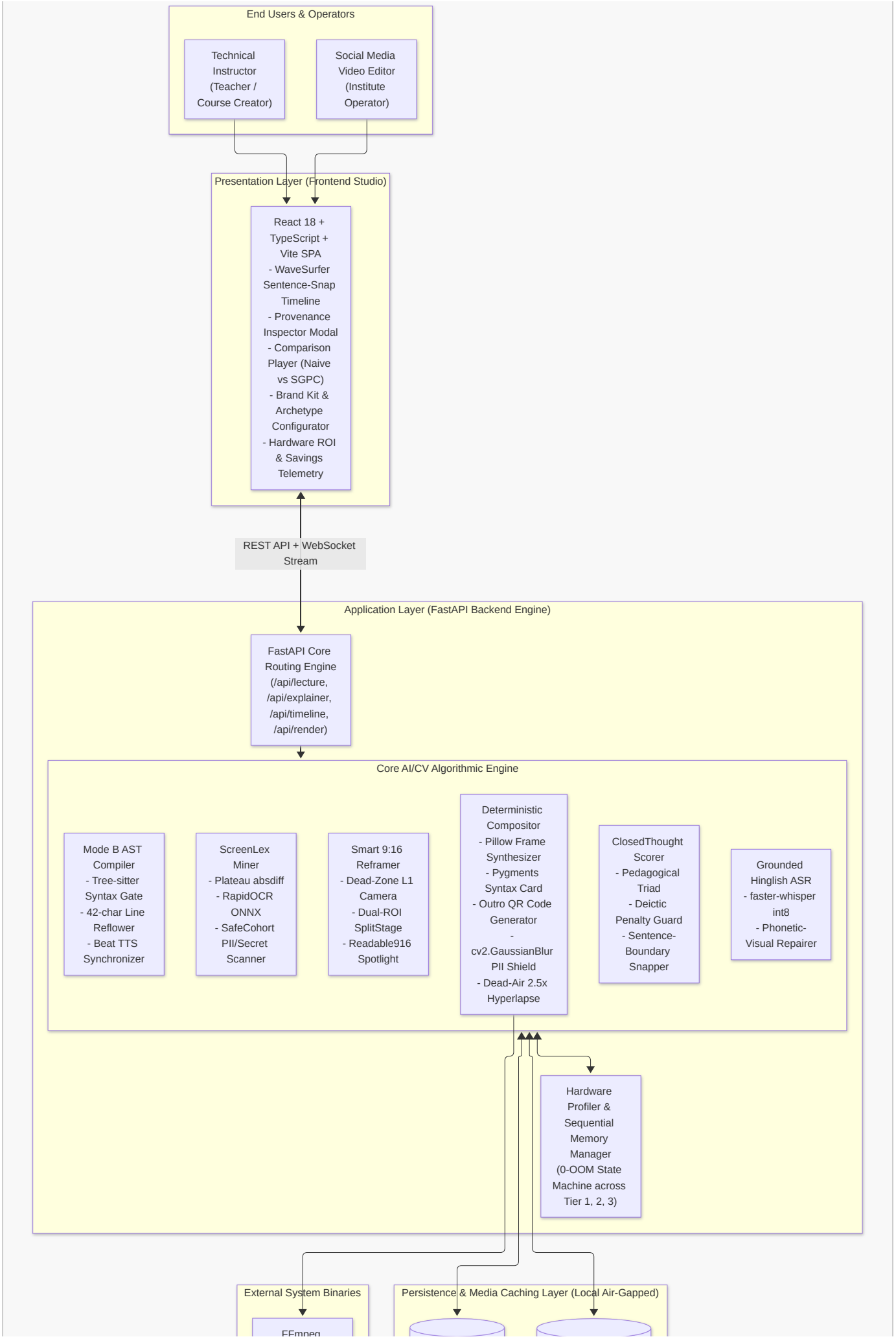
SnipShot's architectural differentiation is structured into **10 modular weapons** designed to dominate technical video creation:

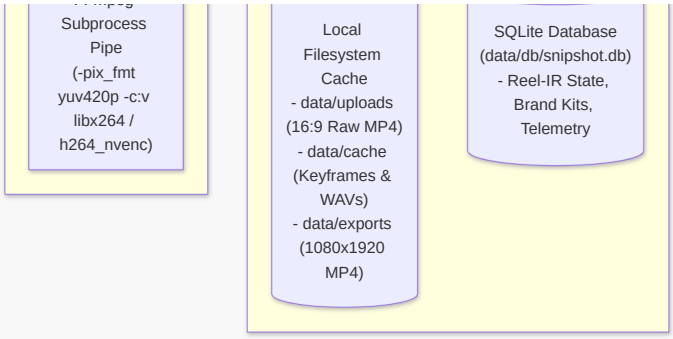
Weapon #	Codename	Core Subsystem	Algorithmic Mechanism	Live Demonstration Proof
1	`ScreenLex-ASR`	`grounded_asr.py` & `phonetic_repair.py`	2-Pass Visual-Lexical Grounding: Plateau RapidOCR tokens bias `faster-whisper` hotwords + Double-Metaphone phonetic repair with Hindi particle guards.	Clickable cyan transcript badges displaying cropped IDE pixel proof for `@Autowired`, `useEffect`, and `kubectl`.
2	`SplitStage`	`smart_reframer.py`	Dual-ROI Transposition: Detects bimodal horizontal activity ($x < 1000$ and $x > 950$), generating an upper $1080 \times 1020\text{px}$ IDE crop and lower $1080 \times 680\text{px}$ Terminal crop.	Instant layout transposition showing live editor typing on top and terminal compilation logs on bottom.
3	`Readable916`	`smart_reframer.py` & `compositor.py`	AST Code Spotlight Card: Code lines $> 850\text{px}$ trigger an in-memory high-contrast 32px monospace overlay card with syntax highlighting.	Side-by-side comparison showing zero characters clipped horizontally on 9:16 mobile screens.
4	`ClosedThought`	`closed_thought.py`	Pedagogical Triad Scorer: $w_1\text{Prosody} + w_2\text{Visual} + w_3\text{Tech} + w_4\text{Triad} -$	Clip starts cleanly on an interrogative hook (*"Why does this*

Weapon #	Codename	Core Subsystem	Algorithmic Mechanism	Live Demonstration Proof
			$w_5\text{Deictic}$ with magnetic sentence-boundary snapping.	*component...*") instead of a dangling pronoun (*"Toh isko..."*).
5	`SafeCohort`	`screen_lex_ocr.py` & `compositor.py`	Secret & PII Blur Shield: Autonomous regex and Shannon entropy inspection on OCR boxes (`sk-`, AWS keys, passwords, emails), applying localized `cv2.GaussianBlur`.	Studio UI toggles real-time blur overlay over `.env` database passwords and exposed API keys in live code.
6	`VerifiedCodeReel`	`tree_sitter_gate.py` & `mode_b_compiler.py`	AST Syntax Gate: Parses snippets with `tree-sitter` to guarantee 0 syntax errors, enforcing a max 42-char line width and frame-synchronized typing per beat.	Generating a 45s Mode B explainer for `"JSON vs XML"` with validated syntax and synchronized neural voiceover.
7	`ArchetypePacks`	`compositor.py` & `reel_ir.py`	4 Pedagogical Templates: (1) `standard_spotlight`, (2) `bug_hunter` (header shifts Crimson $\rightarrow$ Emerald at fix), (3) `vs_battle` (split comparison), (4) `predict_output` (pause callout).	Ingesting a debugging clip and watching the header banner dynamically turn Emerald Green when the unit test passes.
8	`SocialGrowthPack`	`screen_lex_ocr.py` & `compositor.py`	Screen-Grounded SEO + Outro QR: Generates Broad/Stack/Micro hashtags from AST tokens and renders an offline 'Scan-to-Copy' QR code on the outro.	Scanning the rendered Reel on a mobile phone with a QR scanner to immediately copy the exact code snippet.
9	`TypingHyperlapse`	`compositor.py` & `reel_ir.py`	Dead-Air Fast-Forward: Detects silent intervals $> 1.2\text{s}$ where instructor is typing without speaking and fast-forwards at 2.5x speed.	Eliminates 12 seconds of awkward silent typing while preserving full visual verification of code entry.
10	`LocalROI`	`hardware_profiler.py` & `Header.tsx`	Local Economics Telemetry: Auto-detects hardware tier, computes ₹0.00 cloud costs, and displays lifetime SaaS credits saved.	Studio header widget showing: `Tier 2 Intel Core Ultra 5

4. End-to-End Component & Data-Flow Architecture

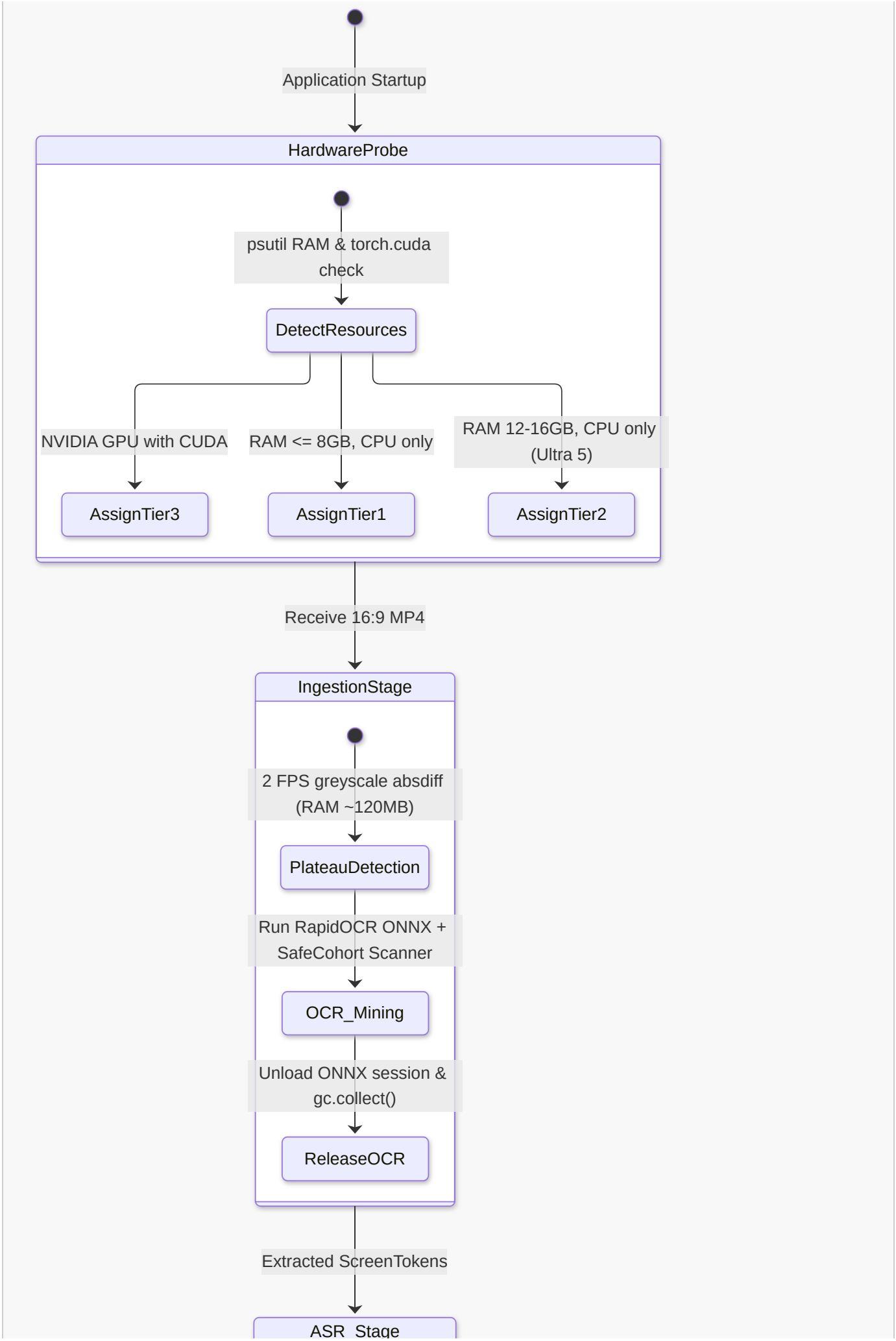
A. C4 System Context & Container Diagram

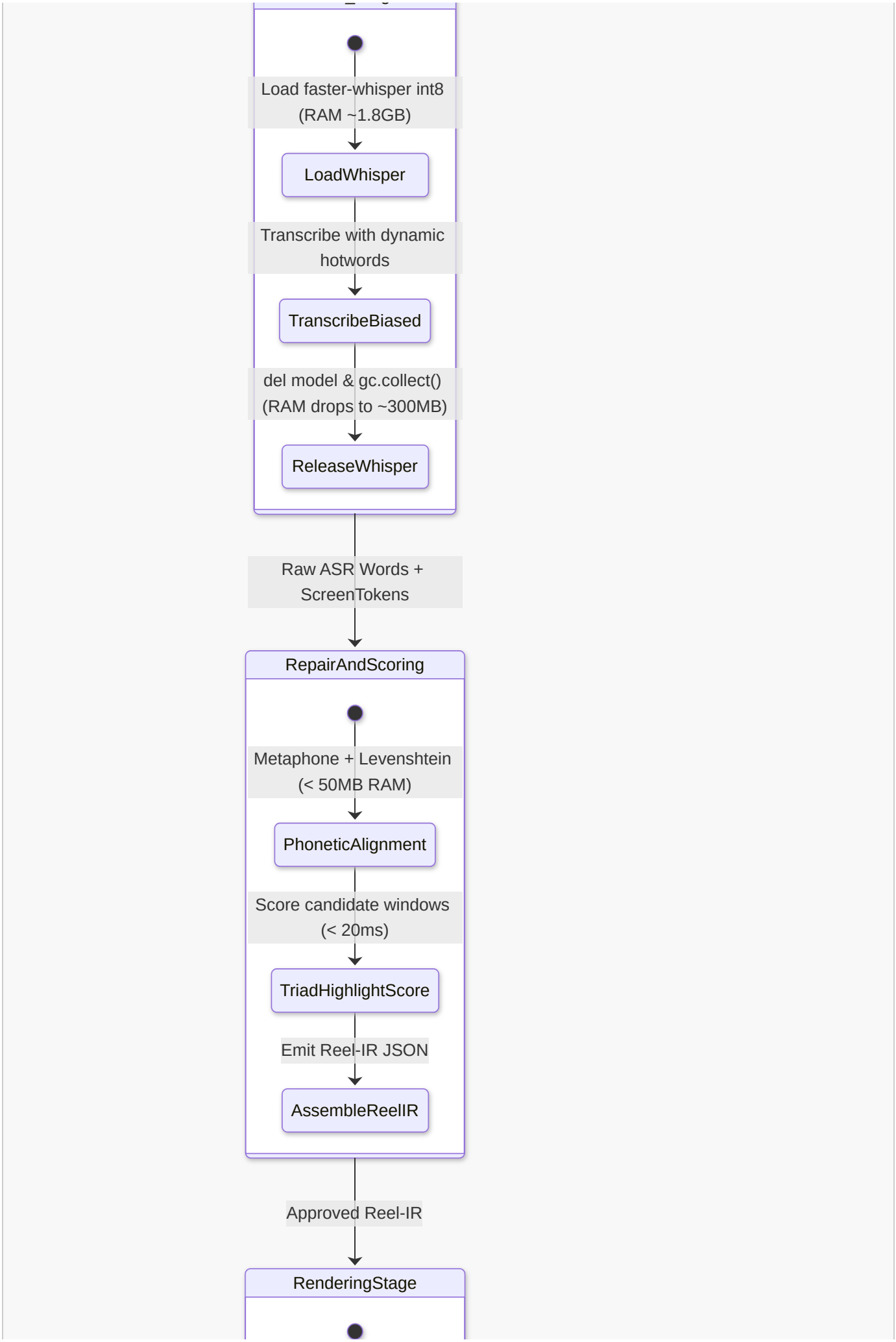

End Users & Operators
Technical Instructor (Teacher / Course Creator)
Social Media Video Editor (Institute Operator)
Presentation Layer (Frontend Studio)
React 18 + TypeScript + Vite SPA
- WaveSurfer Sentence-Snap Timeline
- Provenance Inspector Modal
- Comparison Player (Naive vs SGPC)
- Brand Kit & Archetype Configurator
- Hardware ROI & Savings Telemetry
REST API + WebSocket Stream
Application Layer (FastAPI Backend Engine)
FastAPI Core Routing Engine (/api/lecture, /api/explainer, /api/timeline, /api/render)
Core AI/CV Algorithmic Engine
Mode B AST Compiler
- Tree-sitter Syntax Gate
- 42-char Line Reflower
- Beat TTS Synchronizer
ScreenLex Miner
- Plateau absdiff
- RapidOCR ONNX
- SafeCohort PII/Secret Scanner
Smart 9:16 Reframer
- Dead-Zone L1 Camera
- Dual-ROI SplitStage
- Readable916 Spotlight
Deterministic Compositor
- Pillow Frame Synthesizer
- Pygments Syntax Card
- Outro QR Code Generator
- cv2.GaussianBlur PII Shield
- Dead-Air 2.5x Hyperlapse
ClosedThought Scorer
- Pedagogical Triad
- Deictic Penalty Guard
- Sentence-Boundary Snapper
Grounded Hinglish ASR
- faster-whisper int8
- Phonetic-Visual Repairer
Hardware Profiler & Sequential Memory Manager (0-OOM State Machine across Tier 1, 2, 3)
External System Binaries
FFmpeg
Persistence & Media Caching Layer (Local Air-Gapped)

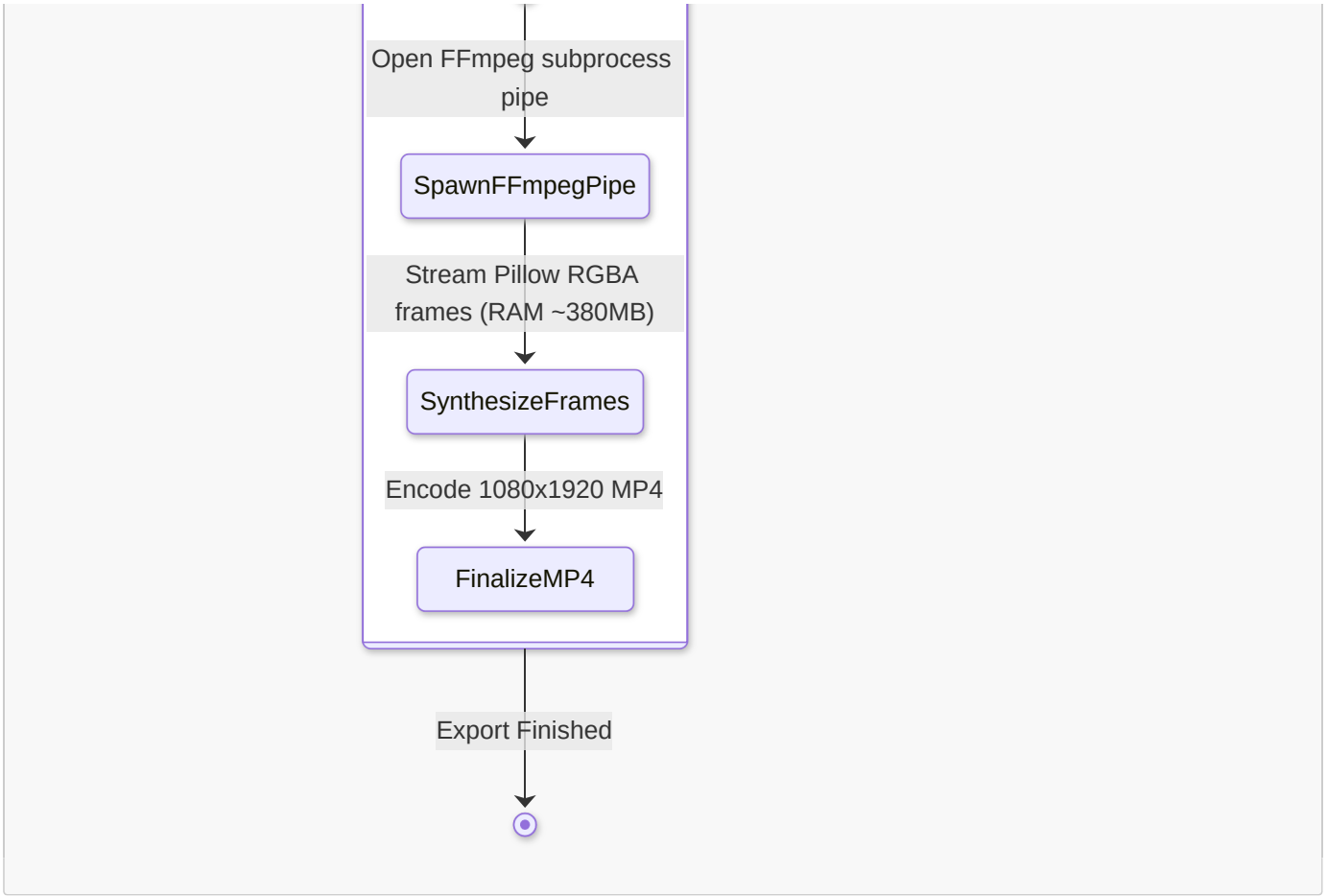


B. Hardware Tier & Sequential Memory State Machine (0-OOM Guarantee)

To guarantee that SnipShot never crashes with an Out-of-Memory (OOM) error on an 8GB or 16GB laptop (such as your Intel Core Ultra 5), models are staged sequentially:


Application Startup
HardwareProbe
psutil RAM & torch.cuda check
DetectResources
NVIDIA GPU with CUDA
RAM <= 8GB, CPU only
RAM 12-16GB, CPU only (Ultra 5)
AssignTier3
AssignTier1
AssignTier2
Receive 16:9 MP4
IngestionStage
2 FPS greyscale absdiff (RAM ~120MB)
PlateauDetection
Run RapidOCR ONNX + SafeCohort Scanner
OCR_Mining
Unload ONNX session & gc.collect()
ReleaseOCR
Extracted ScreenTokens
ASR_Stage


Load faster-whisper int8
(RAM ~1.8GB)
LoadWhisper
Transcribe with dynamic
hotwords
TranscribeBiased
del model & gc.collect()
(RAM drops to ~300MB)
ReleaseWhisper
Raw ASR Words +
ScreenTokens
RepairAndScoring
Metaphone + Levenshtein
(< 50MB RAM)
PhoneticAlignment
Score candidate windows
(< 20ms)
TriadHighlightScore
Emit Reel-IR JSON
AssembleReelIR
Approved Reel-IR
RenderingStage



C. Detailed `Reel-IR` 5-Track + Growth & Redaction Schema Architecture

The `Reel-IR` schema serves as the single source of truth across both Mode A and Mode B:

